

The TikZ-3D Package

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Chapter 1

Introduction

TikZ-3D is a Lua-based 3D graphics package for TikZ.

1.1 What are the current capabilities of TikZ-3D?

TikZ-3D can render multiple non-intersecting triangulated mesh surfaces at once. It is also possible to draw a plane which has been clipped by a cube. There are commands for drawing and labelling geometric vectors too. Curves and surfaces can be defined using matrix transformations.

1.2 How does TikZ-3D improve on existing packages?

There are two significant packages for 3D artwork in TikZ: pgfplots and tikz-3dplot. pgfplots can z-sort the faces of a single parametric surface, but not multiple surfaces at once. Both pgfplots and tikz-3dplot are capable of changing the camera view, but the result is necessarily orthographic, and the rotations are restricted to azimuth and elevation; that is, the z -axis is always vertical. Neither pgfplots nor tikz-3dplot enable matrix math. TikZ-3D is built to render multiple parametric objects at once, enable arbitrary object orientations in non-orthographic projections, and enable matrix math.

1.3 Where is the package at, and where it is heading?

The vision of the package TikZ-3D is to one day be able to render arbitrary numbers of intersecting polygons, by clipping them with each other; this would enable arbitrary surface intersections, without visual artefacts. A more near term goal of TikZ-3D is to also implement an ODE solving tool which approximates trajectories in vector fields.

Chapter 2

Geometric Vectors

2.1



Figure 2.1.1:
The exact location of a geometric vector's tip.

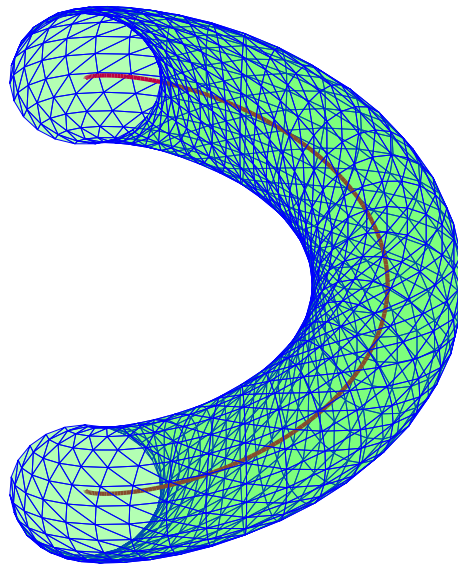


Figure 2.1.2:
A triangulated mesh surface and a curve.